Predicting Diabetes Readmission Rates

Kevin Dao and Taylor Tan

Fall 2026, Erdos Institute Data Science Bootcamp

Importance and Stakeholders

- Diabetes readmissions are common.
  - 1 in 9 hospital encounters are followed by a readmission within 30 days.
- Diabetes readmissions are costly for patients.
- Patients hospitalized with diabetes and their families are the central stakeholders.

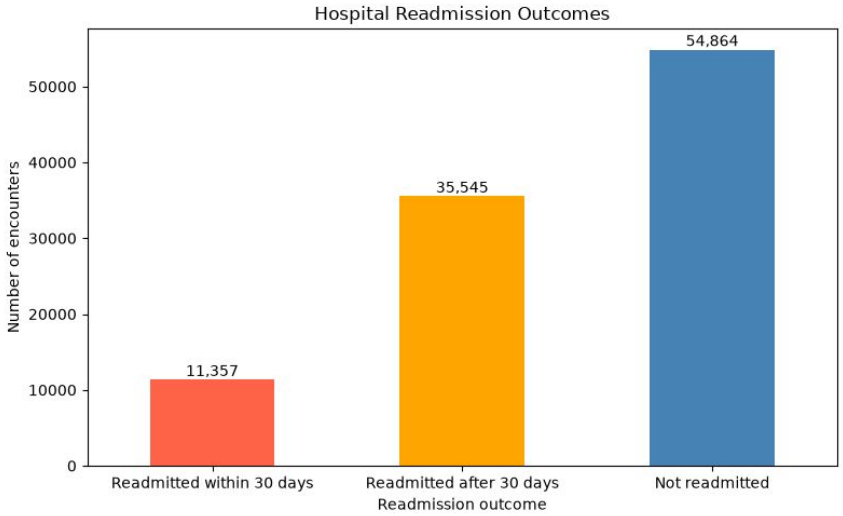
Question: Can we use information available during a hospital stay to identify patients more likely to return within 30 days?

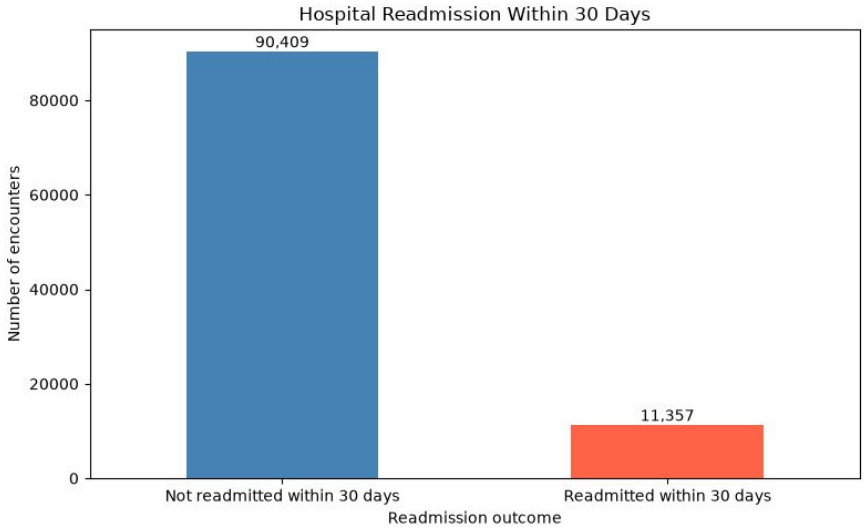
Exploratory Data Analysis
And
Data Cleaning

Diabetes 130-US Hospitals for Years 1999-2008

- Dataset contains 47 Feature Columns.
  - Categorical Features such as Age, Race, Gender.
    - Most categorical features track medications given to patients.
  - Numerical Features such as number of prior hospital visits, number of medications, etc.
- Data set contains 101,766 total visits.
  - 71,518 patients.

First Look at Readmission



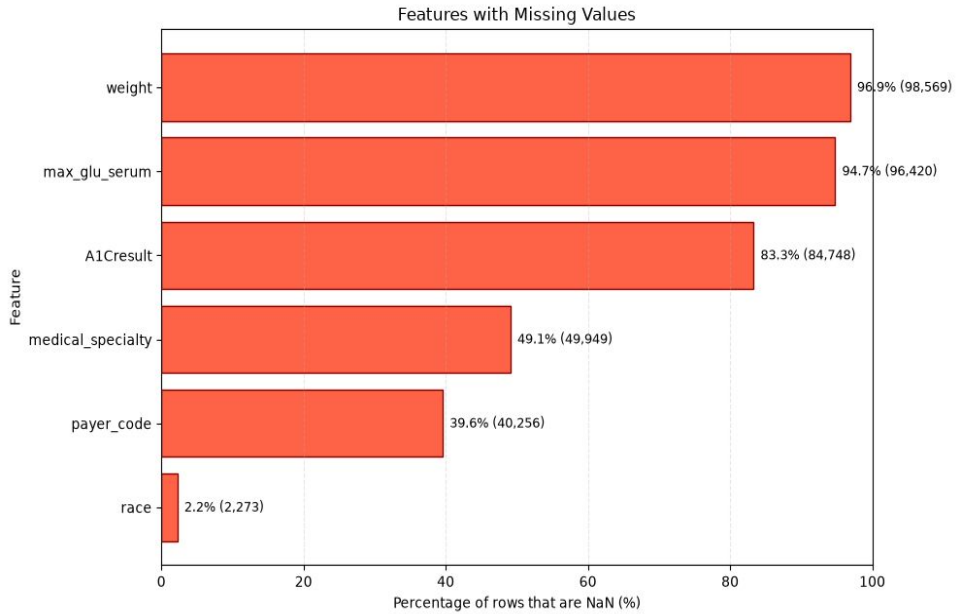


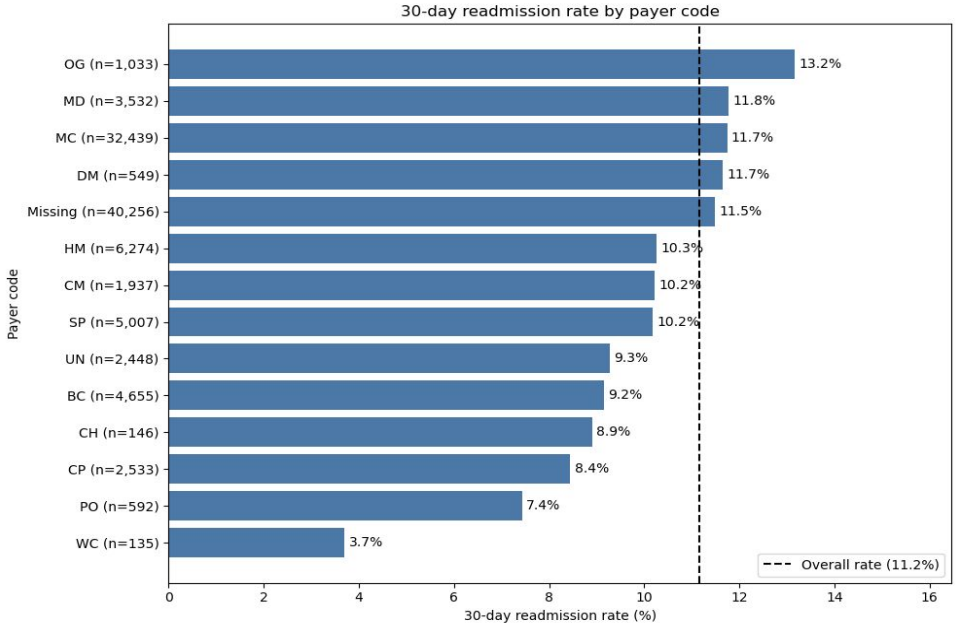
71,518 patients across **130 US hospitals**, 1999–2008.

101,766 total encounters due to repeat patients.

11.16% readmission rate in 30 days.

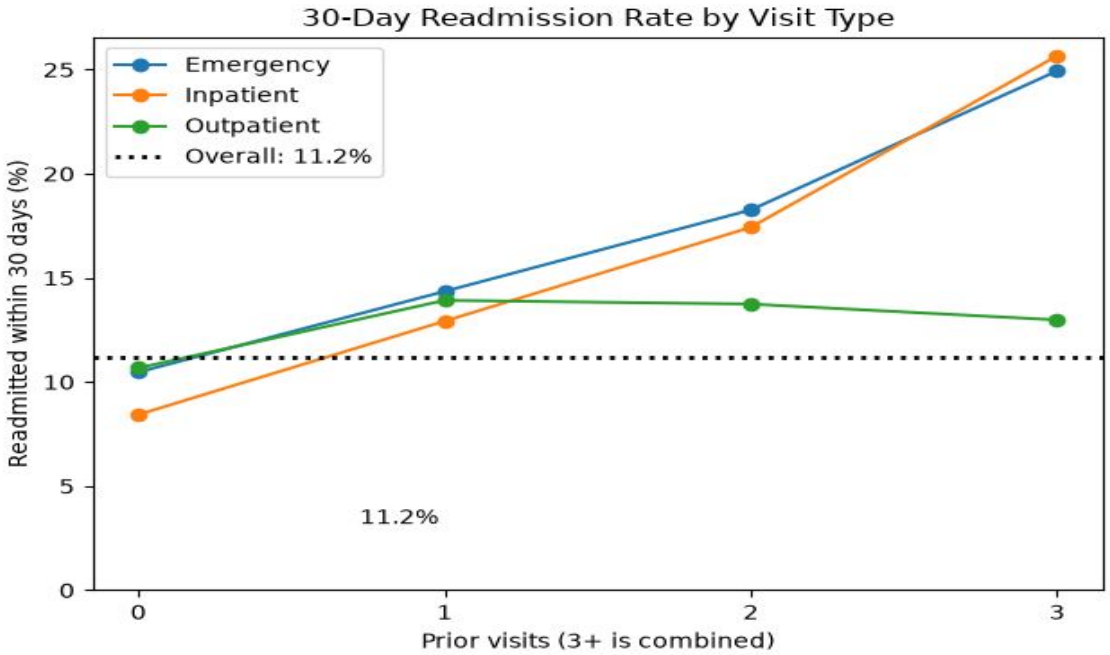
Features with Missing Values





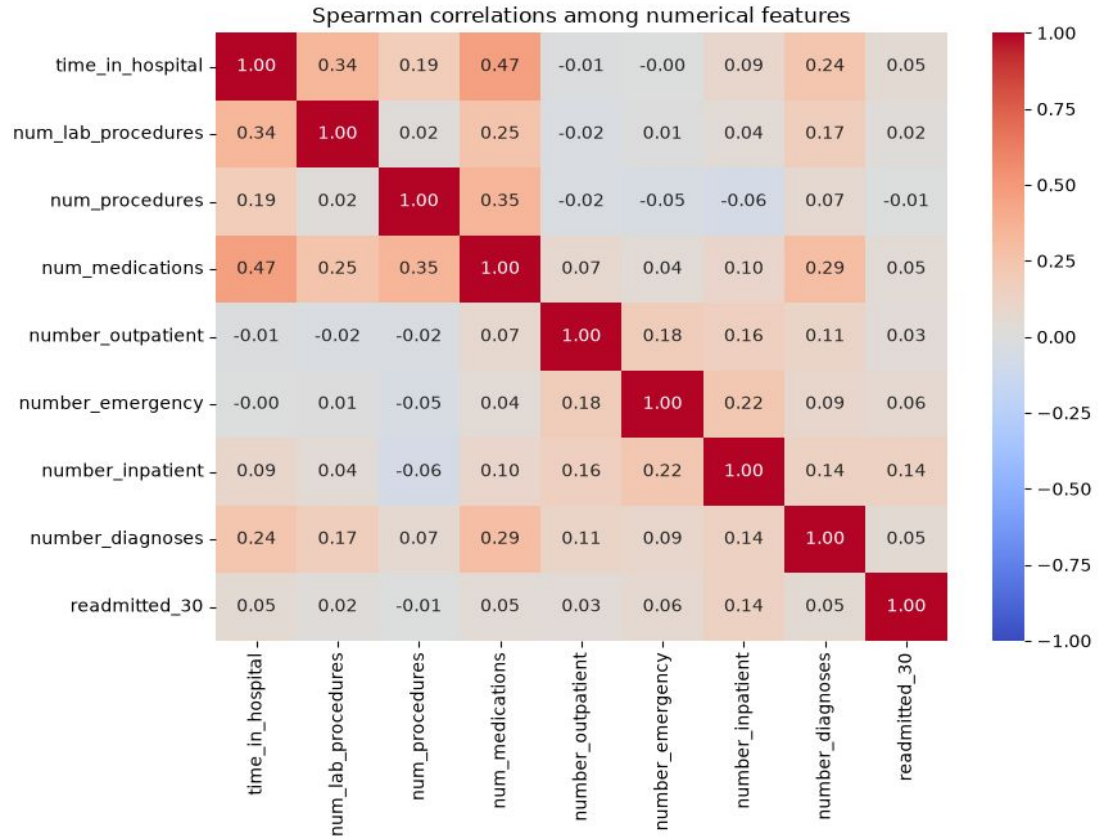
- Payer Code tracks the patient's health insurance.
  - Most patients with entries are on Medicare.
  - Missing entries were still above the overall rate.

Past Visits as a Predictor



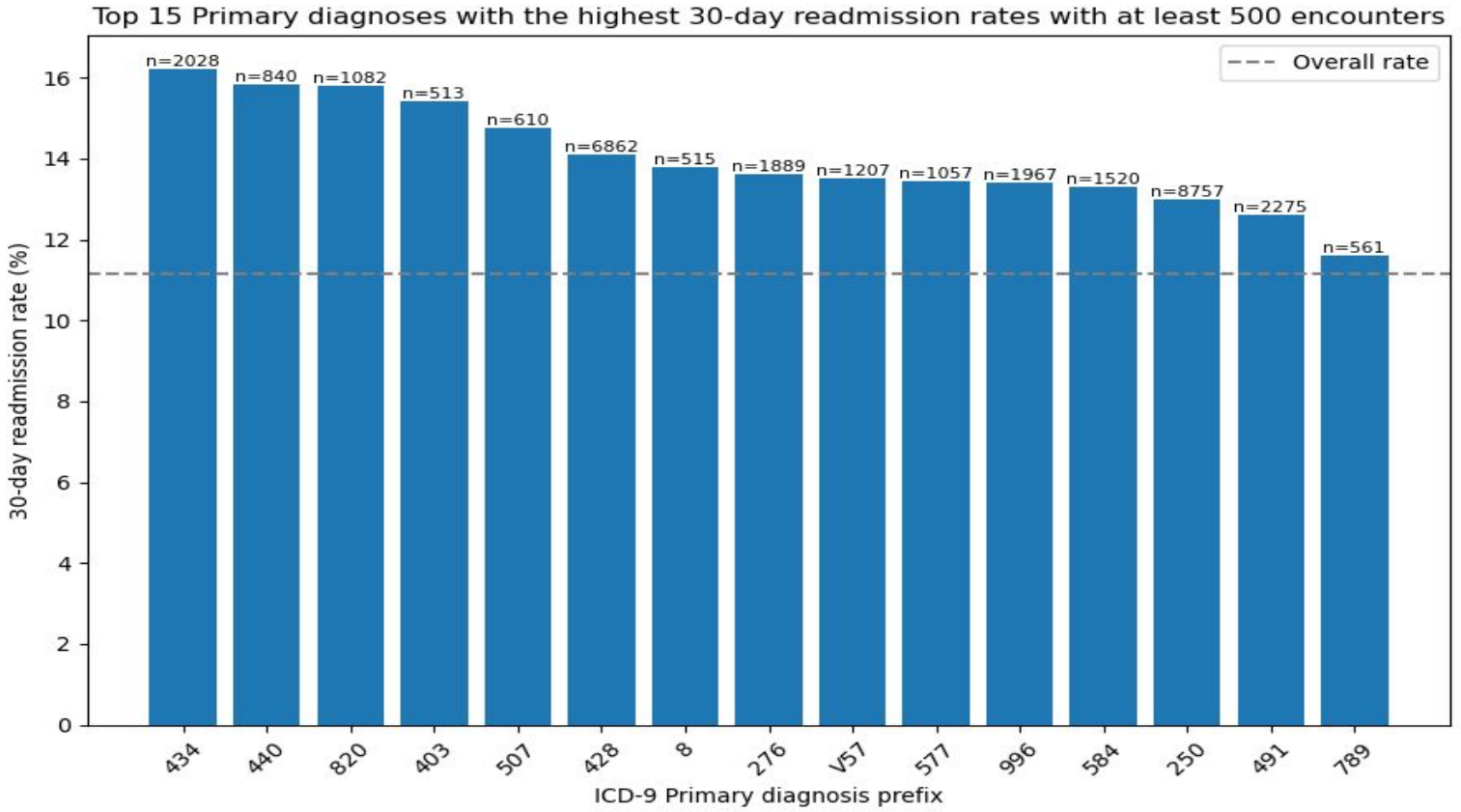
More than 2x the overall readmission rate for patients with history of ER or prior overnight stays.

Spearman Correlations between Numerical Columns



- Most correlations occur with time in hospital.
- Most other pairings had low correlation.

Influence of primary Diagnoses on Readmission Rates



Modeling Process and Methodology

Modeling Process and Experiments I

- **Main KPI: F2 score** $$F_2 = \frac{5 \times precision \times recall}{(4 \times precision) + recall}$$
  - To emphasize recall over precision.
  - Emergency hospitalizations are far more costly for patients.

- **Secondary KPIs: Precision and Recall**
  - Still use precision and recall to gauge performance.
  - Possible to have decent F2-score (F2=0.5)while having not satisfactory precision and recall (both=0.5).

Key points to keep in mind.

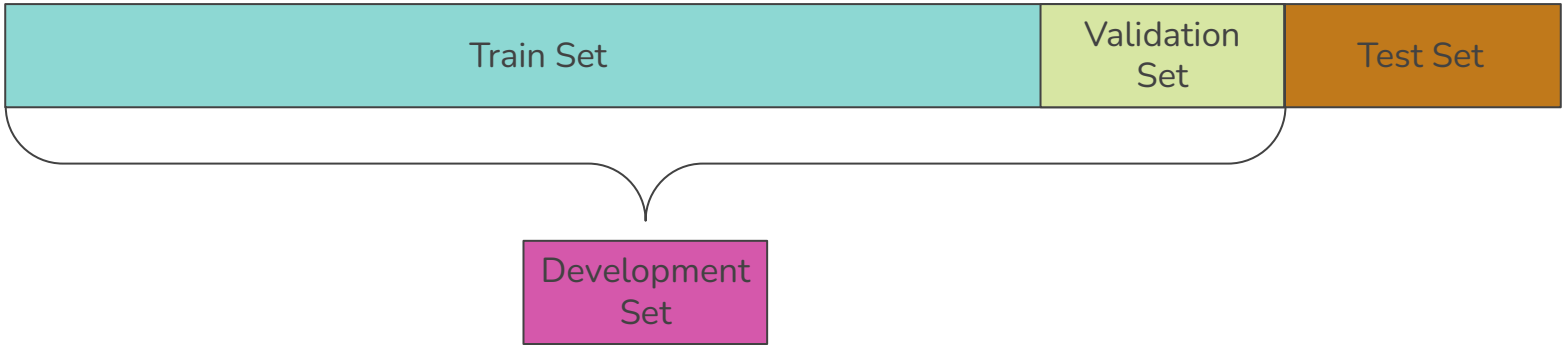
- Data includes diabetes medications (both generation one and two such as sulfonylureas).
- Data tracks dosage in the encounter
  - ➢ Displayed on the right is **repaglinide** and **nateglinide**.
- Data contained 23 such features.
- Point of focus: **can we find a model that uses substantially fewer number of features?**

repaglinide	encounters	readmission_rate_pct
No	100227	11.126742
Steady	1384	13.150289
Up	110	18.181818
Down	45	6.666667

nateglinide	encounters	readmission_rate_pct
No	101063	11.158386
Steady	668	11.676647
Up	24	4.166667
Down	11	9.090909

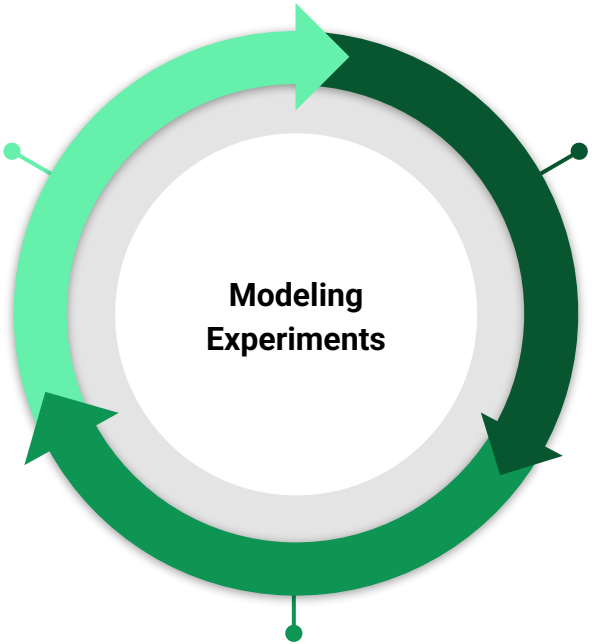
Modeling Process and Experiments II

- First perform an 80/20 split for a development set/test set.
- Further perform 80/20 split of the development set into a train/validation set.



- We should try to keep repeat encounters from the same patient on the same side of a split.
- Preserve the proportions of positive and negative cases.
- For this, we used sklearn's StratifiedGroupKFold

Modeling Process and Experiments III



Consider what the models do well.

Is the tuned threshold capturing noise? What are the most important features? Are the results convincing?

Design models based on insights.

Fit models to the train set.

Use validation set to tune thresholds and compare.

Find thresholds for good F2-score of fitted models.

Modeling Process and Experiments IV

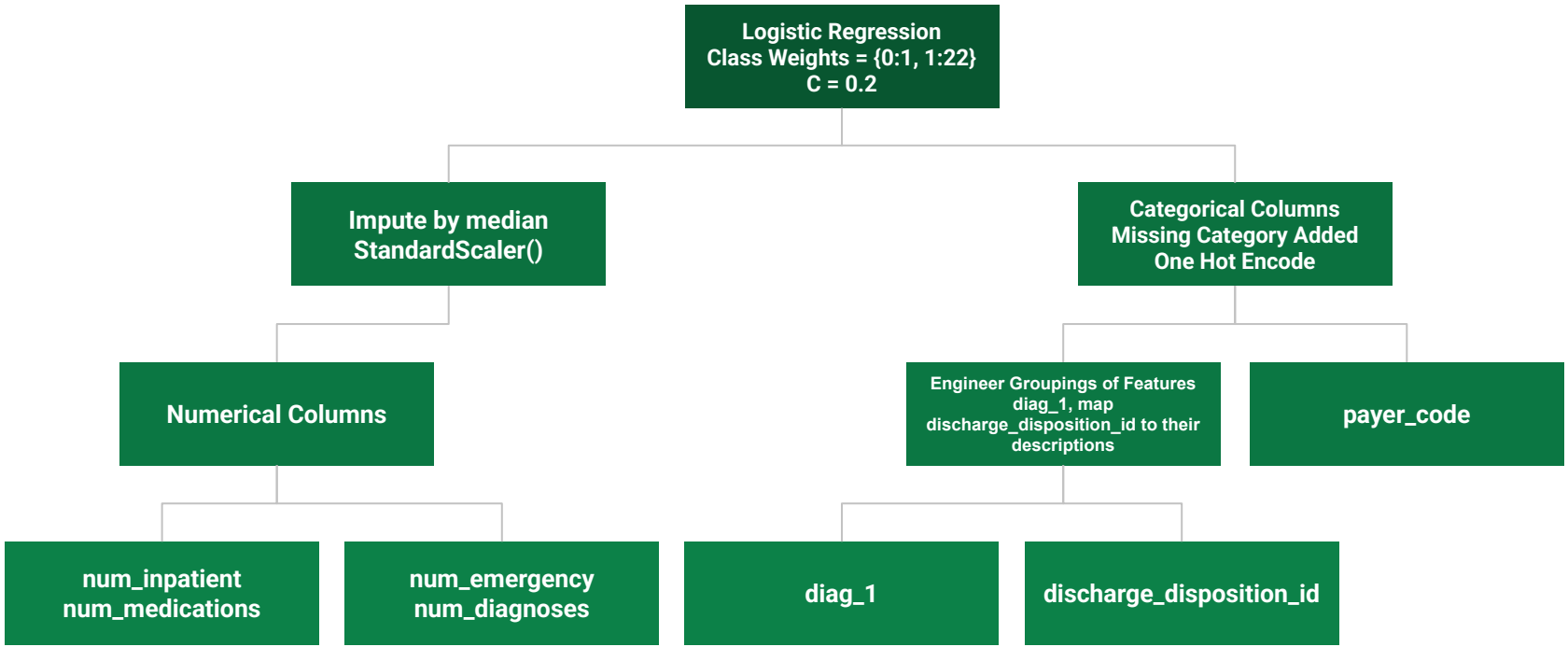
Validation Results

Models:	XGBoost Classifier	Logistic Regression (seven original features chosen with groupings)	Logistic Regression (heavy feature engineering and tuning)	CatBoost	LightGBM	Random Forest Classifier
F_2 Scores:	0.39261 at threshold 0.11102	0.42735 at threshold 0.65989	0.43951 at threshold 0.44835	0.42915 at threshold 0.43757	0.42127 at threshold 0.22042	0.42394 at threshold 0.17200
Precision/Recall:	0.12/0.95	0.15/0.81	0.17/0.74	0.17/0.71	0.14/0.85	0.15/0.79

Evaluating on Test Set

Model	**Baseline**	**Logistic Regression Light**
Description	Plain logistic regression, minimal preprocessing using all features.	Uses features from the original dataset, performed imputations and standard preprocessing, performed groupings for diagnoses
Chosen Threshold p	p=0.06736	p=0.65989
F2-Score on Test	0.41277	0.42477
Precision/Recall on Test	0.13/0.86	0.15/0.81

The Final Logistic Regression Model

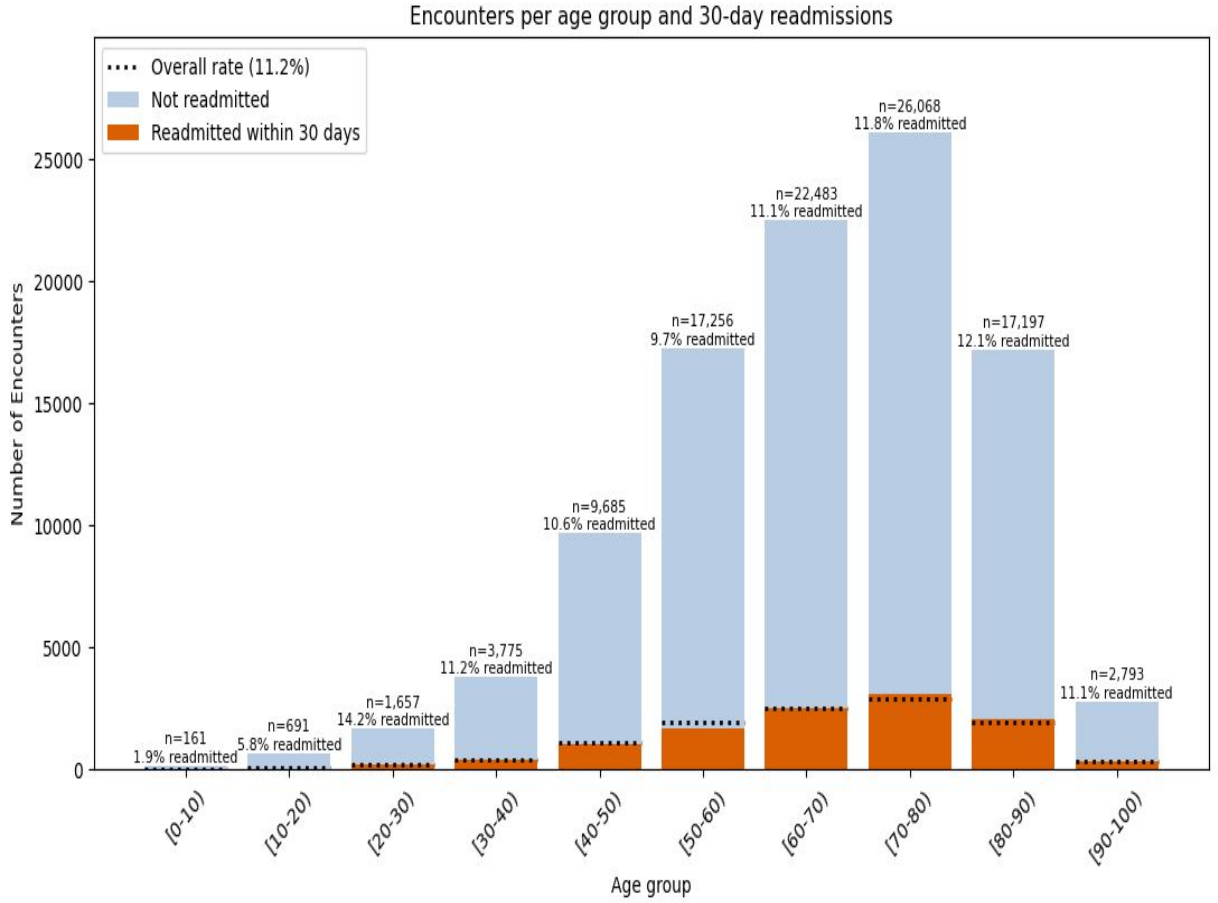


Future Directions

Future Work

Our model did not substantially beat the baseline in F2 score, but its smaller feature set could lead to better models.

- Perform further data collection.
- Explore ensemble models more deeply.
- Focus on subsets of the datasets e.g. separate models for different age groups or those with A1C results.
- Further feature engineering
- Further analysis of model weights needed.
- Rewrite final model's pipeline for easier production use.



Thank you for listening!